DevOps CI/CD with Jenkins
Software installation

1 Compulsory Software

This software is required for the workshop. If you have problems installing, please contact me at:

victor.tan.33@gmail.com

1.1 Proper text editor

You will need a reasonably good text editor.

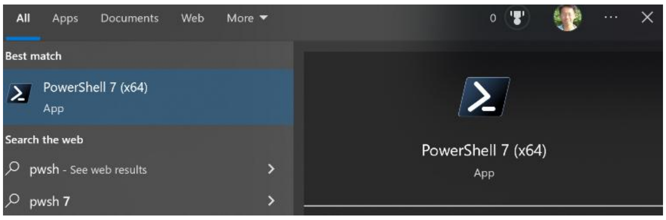
A good option for Windows is NotePad++.

If you have some alternative editor equivalent to NotePad++ which you are familiar and comfortable with, go ahead and use that instead.
Please **DON'T USE** the basic Notepad that comes with all Windows machines: this can cause problems with line formatting and alignment in certain parts of the workshop lab when we are editing material.

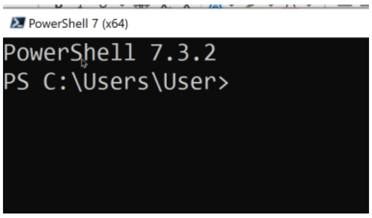
1.2 Powershell 7

You can download and install it from here.

Once installed, search for `pwsh` in the Windows search box at the bottom left, and you should be able to see an icon for this app.

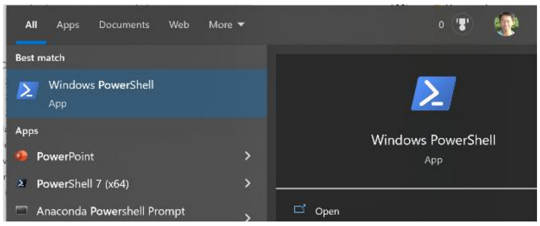


Once it starts up, you should see something like this:

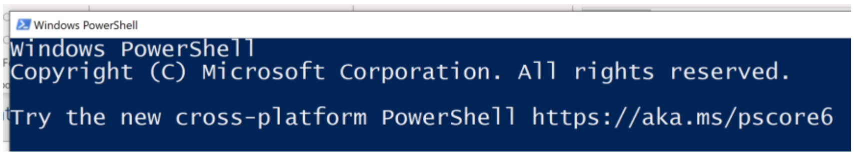


NOTE: This Powershell 7 is a different application from the standard Powershell that comes with all Windows 10 installations - that one is known as Powershell 5.1 or simply Windows Powershell). Powershell 7 is a cross platform version (meaning it can run on BOTH Windows and Linux) of the original Powershell 5.1.

Installing Powershell 7 will NOT OVERWRITE Powershell 5.1, you can still access Powershell 5.1 by typing `powershell` in the Search box:



When it starts up, the shell will look like this:



1.3 MobaXTerm (SSH / SCP Windows client)

This is for establishing remote connection to Linux machines. If you already have another client that you frequently use (such as Putty) for this purpose and you are familiar with how to use that, then you don’t have to install this.

Install and download from here. Make sure you download the **(Installer Edition) - GREEN BUTTON**

1.4 Sign up for a Gmail account (if you don't have one)

NOTE: The reason we need a Gmail account for this workshop is because we are going to use Google's free SMTP server service. Therefore, you **CANNOT** use your company / organization email account or Outlook / Yahoo / other provider email account if you have one. You **MUST** create a Gmail account, if you don't already have one. The two links below detail how to do this.

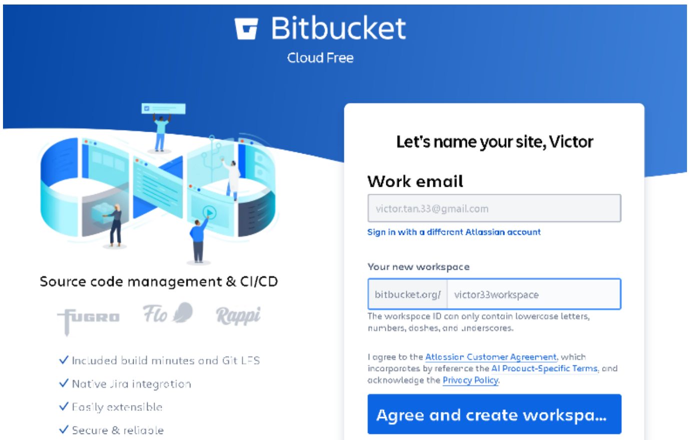
https://www.rightinbox.com/blog/how-to-create-a-new-gmail-account

https://support.google.com/mail/answer/56256?hl=en

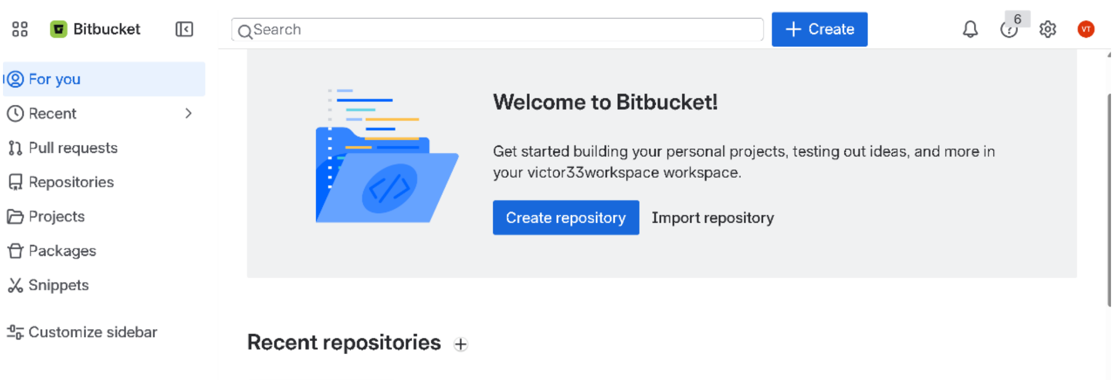
1.5 Sign up for a BitBucket account (if you don't have one)

At the main BitBucket page, you can click on Get It Free, after which you will be presented with an option to either login using any of your existing valid social media accounts, or optionally to create a completely new account on BitBucket.

Once you have logged in successfully, you may need to create a new workspace with a unique name to place all your repositories in.



Once you are done with that, you should be able to see the main page for that workspace.



Many videos on how to do this, this is an example:

https://www.youtube.com/watch?v=IoAqM1CSUHc

1.6 Sign up for a GitHub account (if you don't have one)

At the main GitHub page, you can click on Sign up, after which you will go through the process of creating a new Git Hub account.

Many videos on how to do this, this is an example:
https://www.youtube.com/watch?v=rflDMd5bkz4

2 Optional software installation on Windows

The workshop labs will primarily be conducted on Linux VMs (EC2 instances) running in AWS cloud. This allows me to ensure that all required software is properly configured on these instances, and thereby avoid me having to waste a significant amount of the workshop time in troubleshooting the various software installations on your respective machines due to the unique Windows configuration settings on each of your machines.

However, some of you may wish to run some of the workshop lab exercises on your own machines, in addition to the cloud VMs. The software packages and their installation instructions are therefore provided below. However, you are responsible for troubleshooting and configuring this software to ensure that they run properly

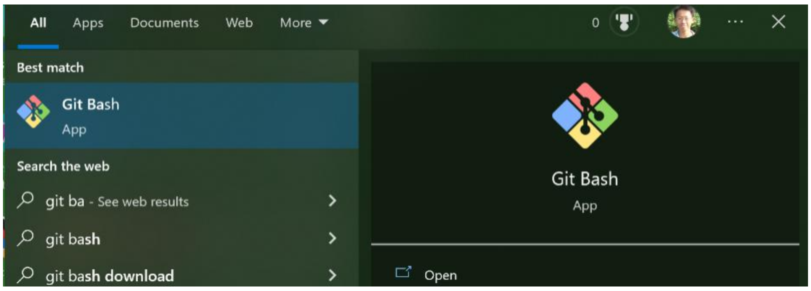
The installation instructions below assume you are using a Windows machine and that have full admin privilege on the machine you are performing installation on. If you are using a company laptop where your account does not have full admin privilege, then you will need to get your system admin to help perform the installation on the company laptop on your behalf.

2.1 Git

Please download and install from this link as it also already includes a BASH shell and GUI as well that we need for working with Git in Windows.

DO NOT use the official Git site because it does not include these additional tools which we need.

Once installed, search for `Git Bash` in the Windows search box at the bottom left, and you should be able to see an icon for this app.



When you start it, the Git Bash shell should look something like this:

If you cannot start the Git Bash shell as shown above, uninstall and repeat the installation process

2.2 Java 21 (or later)

NOTE: You must have **at least Java 21** installed as this is the **minimum requirement for the latest version of Jenkins**.

To determine and install the latest version of Java:

There are several instructions guides for installing Java on Windows as listed below

Please download and install Java 21 / 24 (instead of the edition shown in the instruction guides below). The rest of the steps are identical. Remember to set your `JAVA_HOME` environment variable and include it in the Path variable with `%JAVA_HOME%\bin`

https://devwithus.com/install-java-windows-10/
https://mkyong.com/java/how-to-set-java_home-on-windows-10/
https://www.guru99.com/install-java.html

2.3 Jenkins

The official reference for installing Jenkins on Windows.

An online tutorial that is similar to this.

Some key points to note for the installation process:

Install all suggested plugins during the installation phase

Make sure you note down the admin username and password for accessing Jenkins: keep it simple, for e.g `admin`

When installing using the Windows MSI file, for the Service Logon Credentials prompt, to simplify the setup choose the Logon Type as **Run the service as LocalSystem**. Although this is not recommended from a security viewpoint, this is easier to setup than the other option detailed below and is adequate for the purposes of experimenting with Jenkins on your own machine.

If you wish to use the more secure and recommended option of running the service as a local or domain user, you will probably need to use a local user account (preferably with admin privilege) and enable service logon for that user account.

- Creating a local user account and changing it to admin account
- Enabling service logon

You will also need to be able to access the home directory of the user account that you specified when going through the post installation set up wizard in order to unlock Jenkins.

2.4 Maven

Maven can be downloaded from the official site:

The official reference for installing on Windows

Additional useful installation guides to supplement the reference:

https://mkyong.com/maven/how-to-install-maven-in-windows/
https://howtodoinjava.com/maven/how-to-install-maven-on-windows/
https://stackoverflow.com/questions/17136324/what-is-the-difference-between-m2-home-and-maven-home

2.5 Tomcat

IMPORTANT NOTE FOR PARTICIPANTS WHO DO NOT HAVE ADMIN PRIVILEGE TO THEIR MACHINES DURING THE WORKSHOP PERIOD

We will be modifying certain configuration settings for Tomcat during the workshop. If your administrator installs Tomcat for your machine somewhere inside `C:\ProgramFiles` (the default installation location), you will not be able to access this installation directory without admin privilege. Therefore, you will not be able to perform that lab session properly. To avoid this issue, either

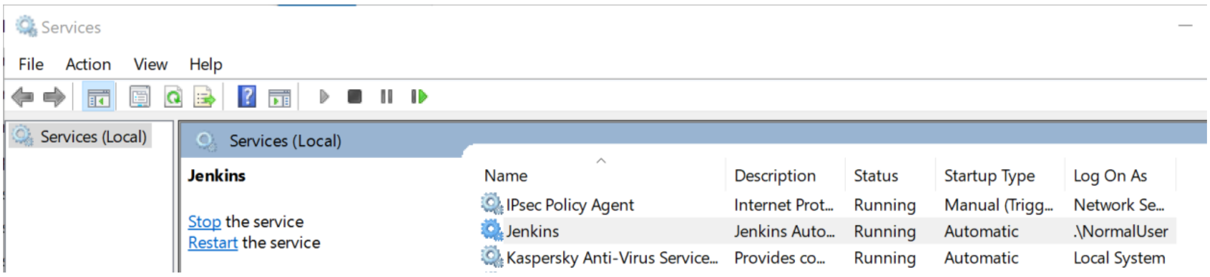
OPTION 1: Install Tomcat yourself in a location that your user account has access to, for e.g.

`C:\Users\`*`your-username`*`\Desktop\Tomcat 9.0`
This means you must explicitly override the default suggested option provided during the installation process, for e.g. `C:\Program Files\Apache Software Foundation\Tomcat 9.0`

OPTION 2: Ask your system admin to give you temporary system admin privilege to the laptop for the duration of the workshop when we are working with Tomcat.

Before installation:

Both the Jenkins Server and the Tomcat server are installed by default to listen on port 8080 (these default ports can be reconfigured later). Before installing Tomcat, **turn off the Jenkins service you installed earlier to avoid port conflicts** during the installation process via the Services app (accessible from Windows Search).



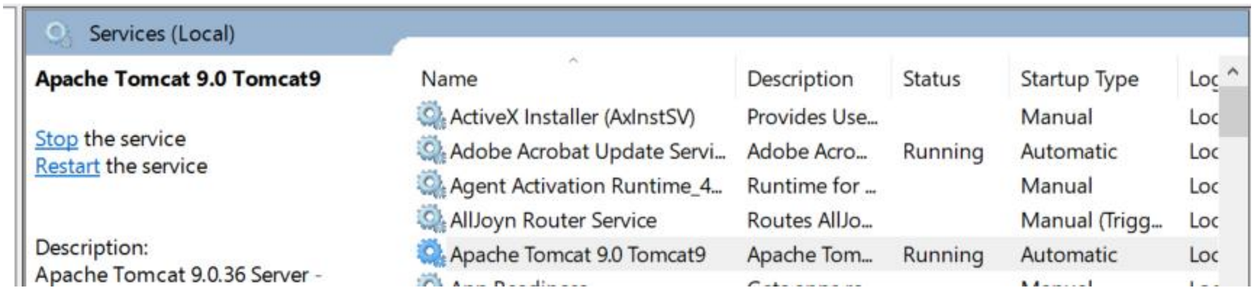
The [latest version of Tomcat for Windows](#) can be downloaded from here.
For a Windows installation, choose the 32-bit/64-bit Windows Service Installer (`apache-tomcat-[version].exe`) - which installs Tomcat as a Windows service

A [complete guide to installing Tomcat](#) on Windows (note that this references version 10, while the latest version is currently higher -> so use the latest version instead).

During the installation process using a Windows installer mode:

- In Choose components, select Full. Make sure the Service start up is checked enable Tomcat to be installed as a service during the installation process.
- Keep the default ports in the Configuration window (for the newer version of Tomcat, the default shutdown port is listed as -1 which will cause a problem later when configuring Eclipse to start Tomcat; accept this for now but you will need to change it later).

Verify the installation is successful by typing `localhost:8080` in a browser tab. You should see the main landing page for Tomcat.



You can bring up the configuration window for Tomcat by right clicking on the Tomcat icon in the bottom right. You can also perform administration through the Services window. Just like any other standard service you can right click to stop it, pause it and restart it.

2.5.1 Configuring Tomcat

Since both Tomcat and Jenkins server share the same default HTTP port, you can change the default port for Tomcat to avoid conflict.
Another reference for changing the Tomcat port number.

Tomcat has a number of default ports:

- Tomcat admin port: 8005
- HTTP/1.1: 8080
- AJP/1.3: 8009

It will not be able to start properly if any other process is bound or listening on those ports. To check for processes that are bound to any of these ports and terminate them, run the following commands in a command prompt in Windows:

```
netstat -aon | findstr port-number
```

```
TCP    0.0.0.0:port-number    0.0.0.0:0              LISTENING       PID
TCP    [::]:port-number       [::]:0                 LISTENING       PID
```

```
tasklist | findstr PID
```

```
someprogram.exe    PID Console                    11    219,532 K
```

```
taskkill /PID PID /F
```

```
SUCCESS: The process with PID PID has been terminated.
```